CSE 471
Introduction to Artificial Intelligence

YooJung Choi

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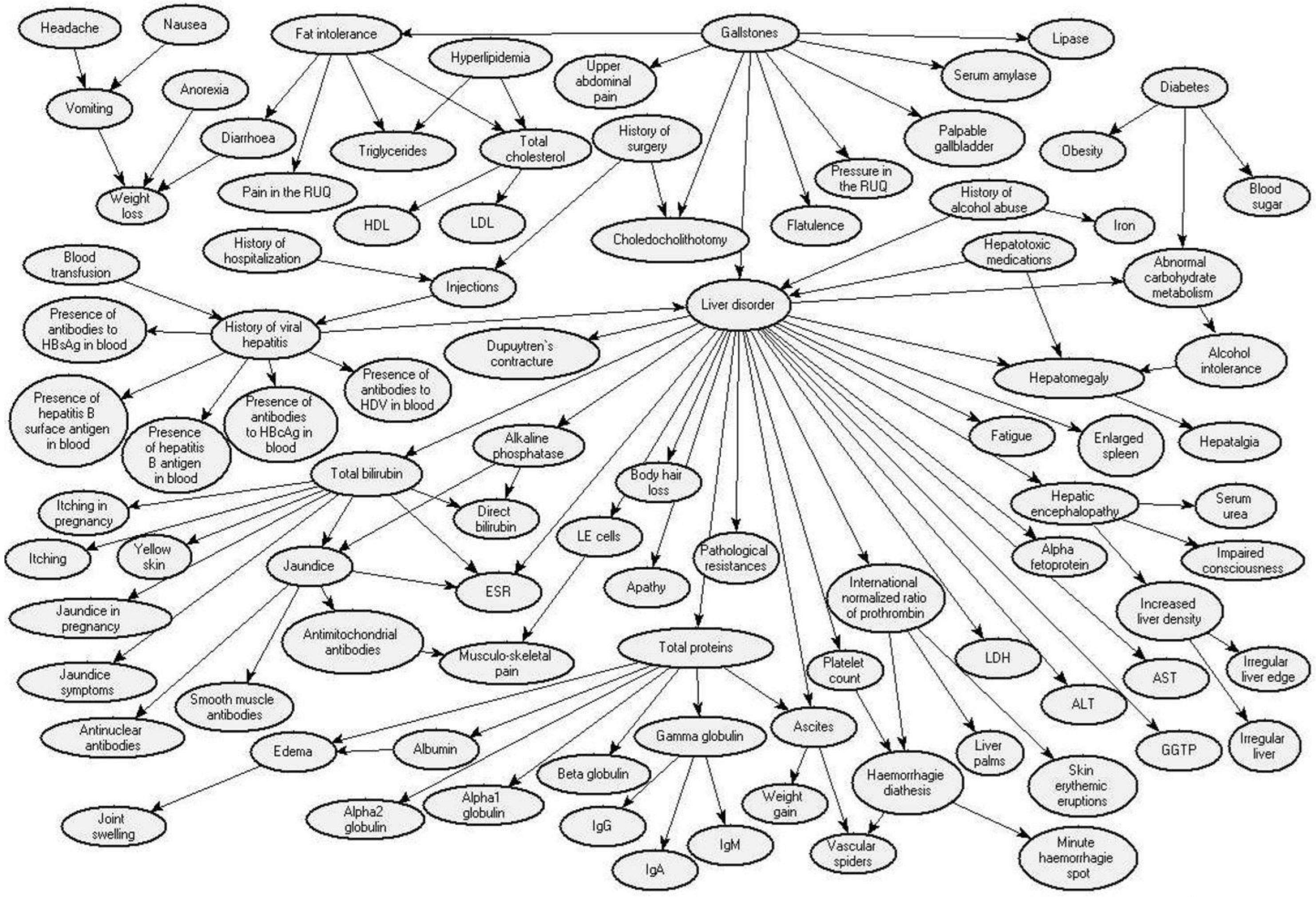
Probabilistic graphical models (PGMs)

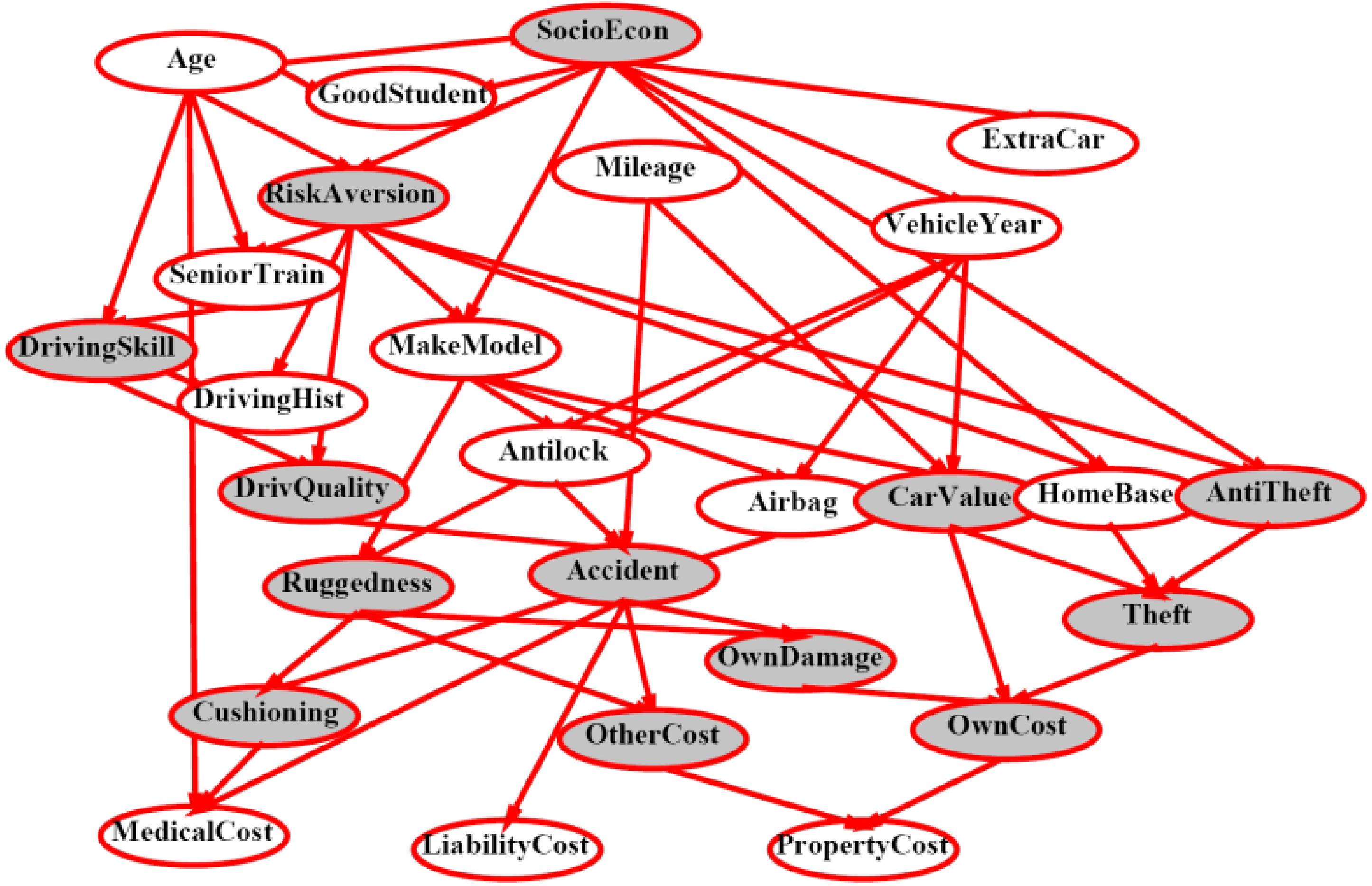
- Two problems with using full joint distribution tables as our probabilistic models:
  - Other than very simple cases, the joint distribution table is too big to represent explicitly
  - Hard to learn (estimate) anything empirically about more than a few variables at a time
- PGMs use a graphical structure to *more concisely represent distributions*
- ***Bayesian networks***: directed PGM exploiting (conditional) independence
  - Describe complex joint distributions using simple, local distributions (conditional probabilities)
  - Local interactions chain together to give global, indirect interactions

BN application: medical diagnosis



A Bayesian Network Model for Diagnosis of Liver Disorders – Agnieszka Onisko, M.S., Marek J. Druzdzel, Ph.D., and Hanna Wasyluk, M.D.,Ph.D.- September 1999.

BN application: insurance



Bayesian network structure

- Nodes: variables (with domains)
  - Can be observed or unobserved
- Arcs: interactions between variables
  - Indicate “direct influence” between variables
  - Formally: encode conditional independence (more later)
- Intuition: it helps to think of arrows as direct causation (though in general, they don’t)

Coin flips

- N independent coin flips

$$X_1 \qquad X_2 \qquad \cdots \qquad X_n$$

- No interaction between variables: *absolute independence*

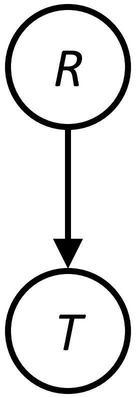
Traffic

- Variables: R: it rains

  T: there is traffic

- Model 1: independence

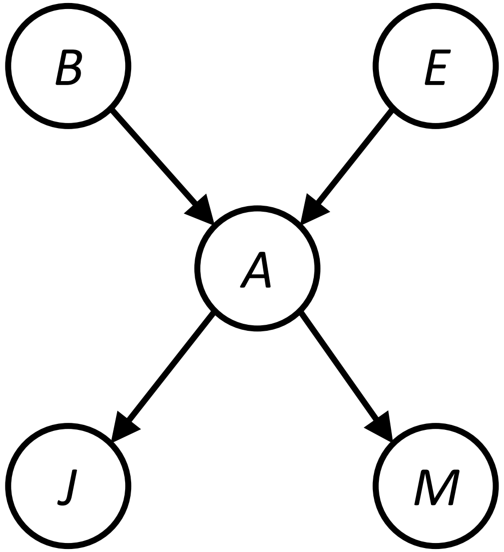
- Model 2: rain causes traffic



- Why is an agent using model 2 better?

Alarm network

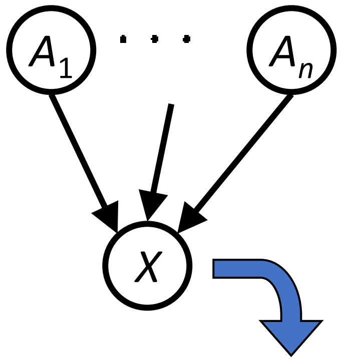
- Variables: B: Burglary
  A: Alarm goes off
  M: Mary calls
  J: John calls
  E: Earthquake



- A burglar or an earthquake may set the alarm off
- The alarm can cause neighbor John or Mary to call

Bayesian network semantics

- A set of nodes, one per variable
- A directed acyclic graph
- A conditional distribution for each node
  - P(X | parents(X))
  - CPT: conditional probability table
  - Description of a noisy "causal" process
- *BN = graph (topology) + local conditional probabilities*



$P(X|A_1 \ldots A_n)$

Probabilities in BNs

- Bayesian networks *implicitly* encode joint distributions As a product of local conditional distributions

$$\boxed{P(x_1, \dots, x_n) = \prod_{i=1}^{n} P(x_i | parents(X_i))}$$

- Check: Are we guaranteed that this results in a valid joint distribution?
- ***Chain rule*** (valid for all distributions): Repeated application of product rule
  - $P(x_1, x_2, \dots, x_n) = P(x_n | x_1, \dots, x_{n-1}) P(x_1, \dots, x_{n-1}) = P(x_n | x_1, \dots, x_{n-1}) P(x_{n-1} | x_1, \dots, x_{n-2}) P(x_1, \dots, x_{n-2})$

$$= \cdots = \prod_{i=1}^{n} P(x_i | x_1, \dots, x_{i-1})$$

- *Note: this implies conditional independences:* $P(x_i | x_1, \dots, x_{i-1}) = P(x_i | parents(X_i))$ *(more on this later)*

Coin flips

$P(X_1)$

h	0.5
t	0.5

$P(X_2)$

h	0.5
t	0.5

$\cdots$

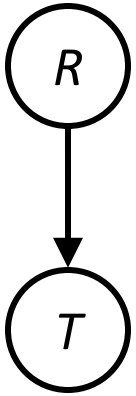
$P(X_n)$

h	0.5
t	0.5

$$P(h,h,t,h) = P(X_1 = h)P(X_2 = h)P(X_3 = t)P(X_4 = h) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{16}$$

Only distributions whose variables are absolutely independent can be represented by a BN with no arcs.

Traffic



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

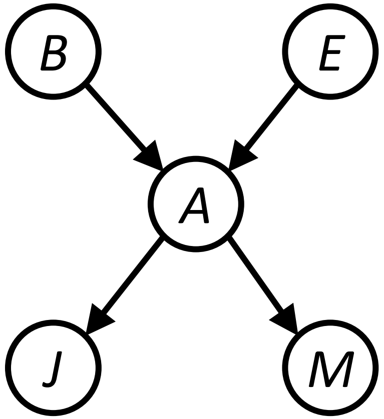
$$P(+r, -t) = P(-t|+r)P(+r) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16}$$

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

Alarm

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m)$$
$$= P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a)$$

Alarm

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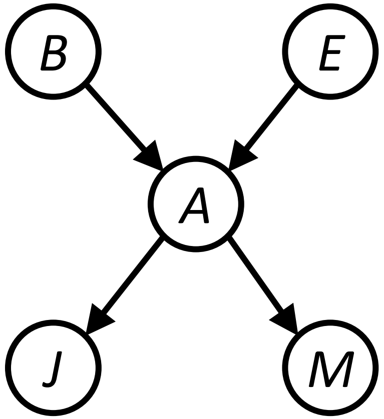

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-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b,-e,+a,-j,+m)$$
$$= P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a)$$
$$= 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

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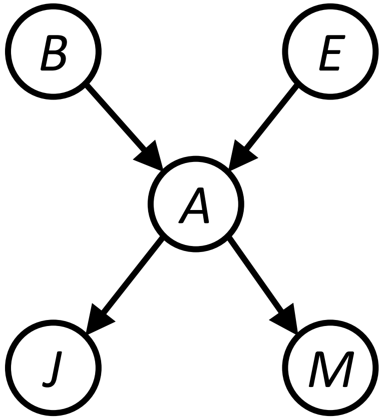

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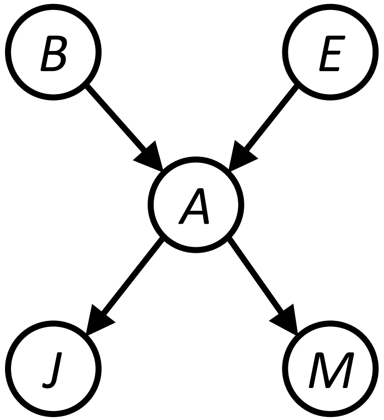
$$= 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

$$P(+b,-e,-a,-j,+m)$$
$$= P(+b)P(-e)P(-a|+b,-e)P(-j|-a)P(+m|-a)$$
$$= 0.001 \times 0.998 \times 0.06 \times 0.95 \times 0.01$$

Size of BN

B	P(B)
+b	0.001
-b	0.999



E	P(E)
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How many independent probability values (free parameters) needed for this table?

A	J	P(J|A)
+a	+j	0.9
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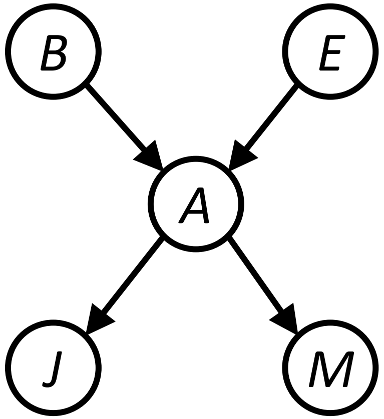

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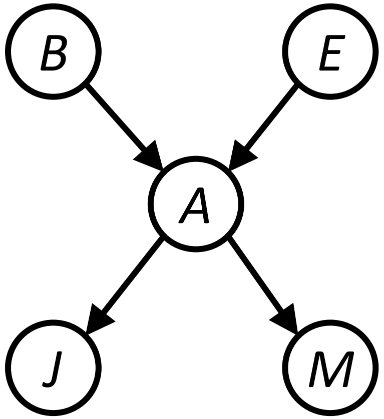

E	P(E)
+e	θ_e
-e	$1-\theta_e$

How many independent probability values (free parameters) needed for this table?

Only 1! The table must sum up to 1.

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
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+a	+m	0.7
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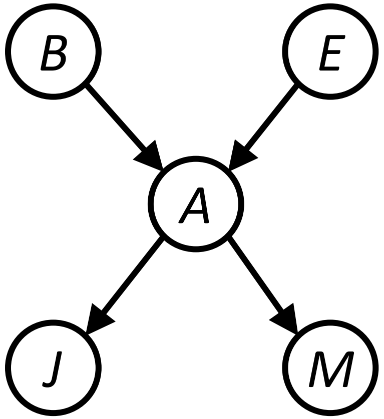

E	P(E)
+e	θ_e
-e	$1-\theta_e$

How many independent probability values (free parameters) in this table?

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
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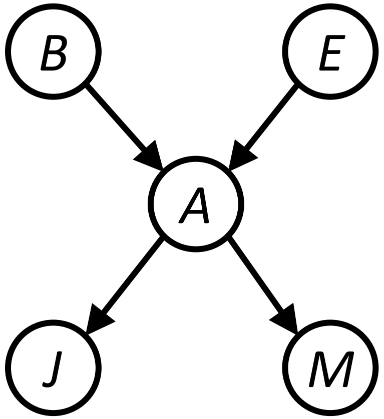

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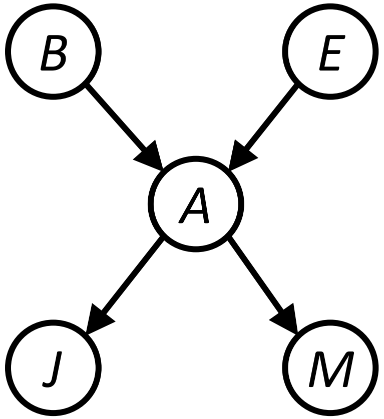
B	E	A	P(A|B,E)
+b	+e	+a	θ_{a1}
+b	+e	-a	$1-\theta_{a1}$
+b	-e	+a	θ_{a2}
+b	-e	-a	$1-\theta_{a2}$
-b	+e	+a	θ_{a3}
-b	+e	-a	$1-\theta_{a3}$
-b	-e	+a	θ_{a4}
-b	-e	-a	$1-\theta_{a4}$

How many independent probability values (free parameters) in this table?

4 free parameters.
For each assignment to the parents, the conditional distribution must sum to 1.

Size of BN

B	P(B)
+b	0.001
-b	0.999



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+e	θ_e
-e	$1-\theta_e$

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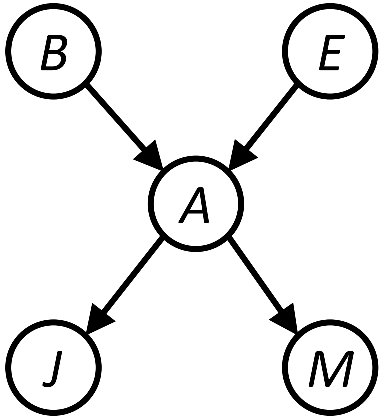
B	E	A	P(A|B,E)
+b	+e	+a	θ_{a1}
+b	+e	-a	$1-\theta_{a1}$
+b	-e	+a	θ_{a2}
+b	-e	-a	$1-\theta_{a2}$
-b	+e	+a	θ_{a3}
-b	+e	-a	$1-\theta_{a3}$
-b	-e	+a	θ_{a4}
-b	-e	-a	$1-\theta_{a4}$

How many independent probability values (free parameters) *for this Bayesian network*?

Size of BN

B	P(B)
+b	θ_b
-b	$1-\theta_b$



E	P(E)
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A	J	P(J|A)
+a	+j	θ_{j1}
+a	-j	$1-\theta_{j1}$
-a	+j	θ_{j2}
-a	-j	$1-\theta_{j2}$

A	M	P(M|A)
+a	+m	θ_{m1}
+a	-m	$1-\theta_{m1}$
-a	+m	θ_{m2}
-a	-m	$1-\theta_{m2}$

B	E	A	P(A|B,E)
+b	+e	+a	θ_{a1}
+b	+e	-a	$1-\theta_{a1}$
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+b	-e	-a	$1-\theta_{a2}$
-b	+e	+a	θ_{a3}
-b	+e	-a	$1-\theta_{a3}$
-b	-e	+a	θ_{a4}
-b	-e	-a	$1-\theta_{a4}$

How many independent probability values (free parameters) *for this Bayesian network*?

1 + 1 + 2 + 2 + 4 = 10

As opposed to $2^5-1=31$ *independent parameters for the full joint distribution table*

Size of BN

- How big is a joint distribution table over N Boolean variables? $O(2^N)$

  over N variables with d values? $O(d^N)$

- How big is a Bayesian network CPT for a Boolean variable that has k parents? $O(2^k)$

  a variable with d values and k parents? $O\left(d^k(d-1)\right) = O(d^{k+1})$

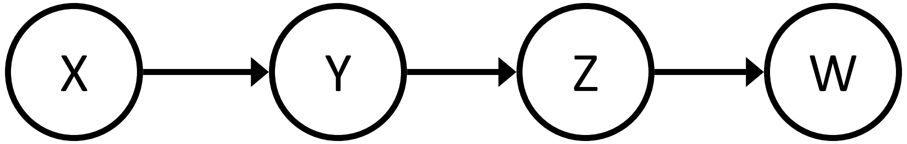
- How big is a Bayesian network with N Boolean variables that have at most k parents? $O\left(N \cdot 2^k\right)$

  with N variables with d values and at most k parents? $O\left(N \cdot d^{k+1}\right)$

- Huge space savings from BNs compared to joint distribution tables!
- Also easier to elicit local CPTs and faster to answer queries

BNs achieve compact representation by making (conditional) independence assumptions!

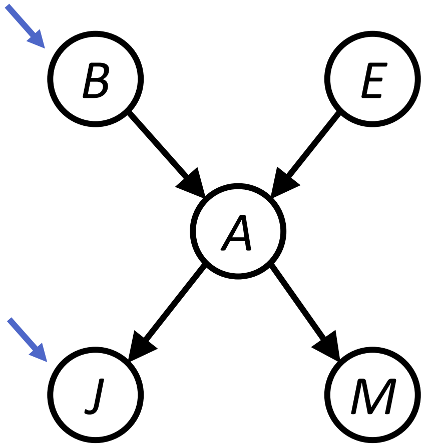
Example: Independence in BN



- Conditional independence assumptions directly from simplifications in chain rule:
  - Chain rule (always true): $P(X,Y,Z,W) = P(X)P(Y|X)P(Z|X,Y)P(W|X,Y,Z)$
  - BN structure implies: $P(X,Y,Z,W) = P(X)P(Y|X)P(Z|Y)P(W|Z)$
- $P(Z|X,Y) = P(Z|Y)$ $\quad$ $Z \perp X \mid Y$
- $P(W|X,Y,Z) = P(W|Z)$ $\quad$ $W \perp X,Y \mid Z$

Independence in BN structure

- Parents(V) = {N: there is an edge N -> V}
- Descendants(V) = {N: there is a directed path from V to N}
- Non-descendants(V): variables other than V, Parents(V), Descendants(V)
- *Markovian assumption*: Every node V is conditionally independent of Non-descendants(V) given Parents(V)



Parents(B) = {}   Non-descendants(B) = {E}

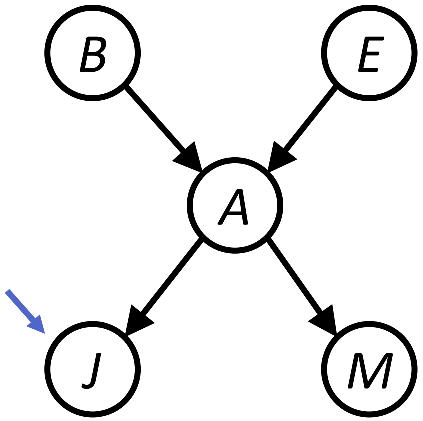
Burglary is independent of Earthquake: $B \perp E$

Parents(J) = {A}   Non-descendants(J) = {B, E, M}

$J \perp B,E,M \mid A$

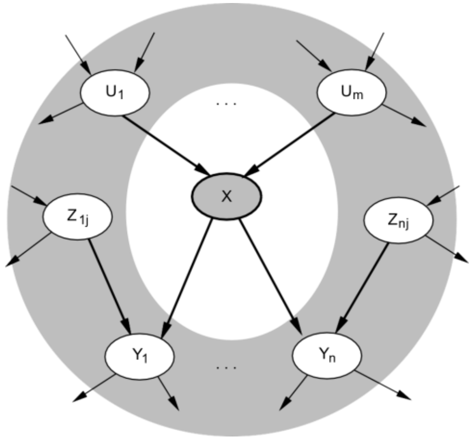
Independence in BN structure

- ***Markov blanket***: parents, children, and children's parents
- Each node is conditionally independent of all other nodes given its Markov blanket



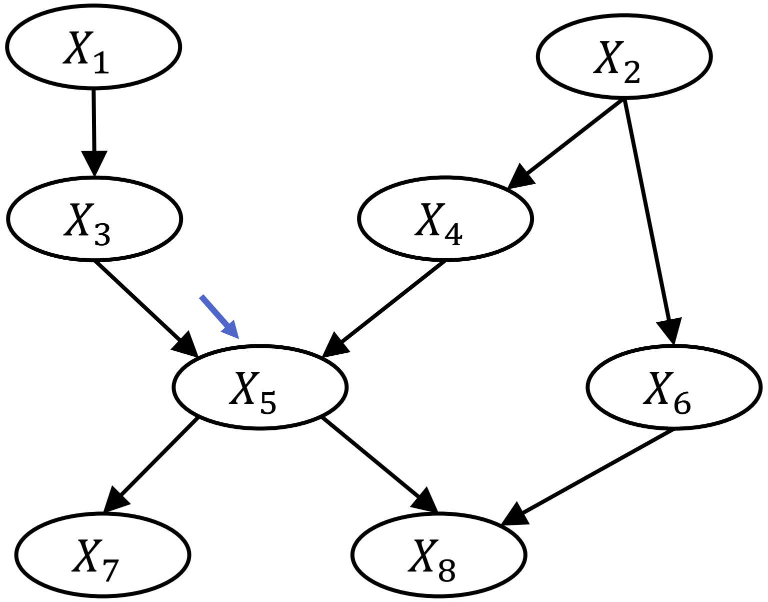
Markov-blanket(J)={A}

$J \perp B,E,M \mid A$



Independence in BN structure

- ***Markov blanket***: parents, children, and children's parents
- Each node is conditionally independent of all other nodes given its Markov blanket



Markov-blanket(X_5) = $\{X_3,X_4,X_6,X_7,X_8\}$

$X_5 \perp X_1,X_2 \mid X_3,X_4,X_6,X_7,X_8$

Parents(X_5) = $\{X_3,X_4\}$

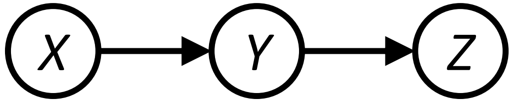
Non-descendants(X_5) = $\{X_1,X_2,X_6\}$

$X_5 \perp X_1,X_2,X_6 \mid X_3,X_4$

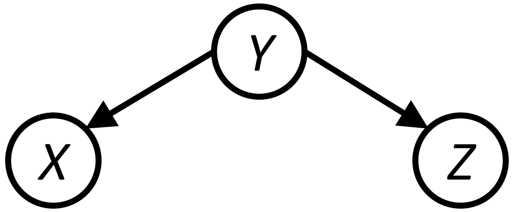
How do we in general prove or disprove whether two nodes are independent given certain evidence?

D-separation

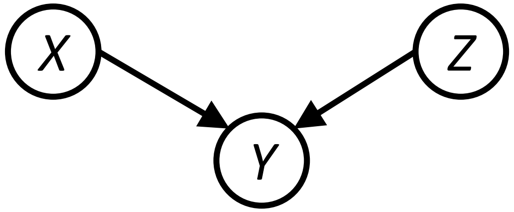
- We can study the (conditional) independence properties assumed by a BN graph structure
- Complex cases can be analyzed using three configurations over triples of nodes:



"Causal chains"

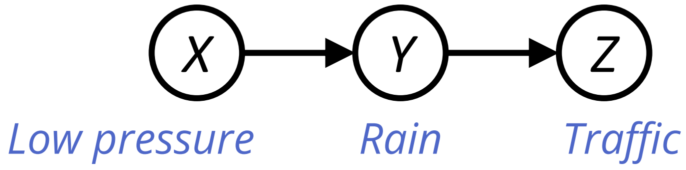


"Common cause"



"Common effect"

Causal chains

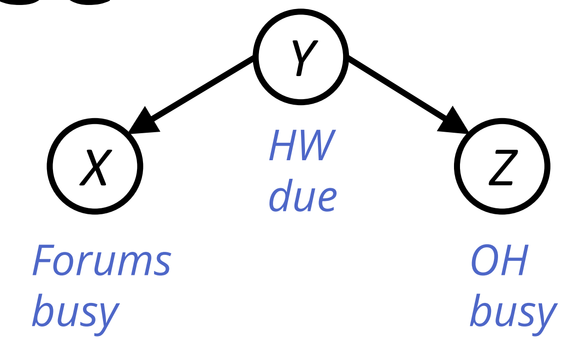


- Is X independent of Z?
  - ***No!*** Suffices to show one counterexample set of CPTs for which X is not independent of Z
  - E.g., low pressure causes rain causes traffic & high pressure causes no rain causes no traffic
- Is X independent of Z given Y? ***Yes!***

$$P(z|x,y) = \frac{P(x,y,z)}{P(x,y)} = \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)} = P(z|y)$$

Evidence along the chain "blocks" the influence

Common cause

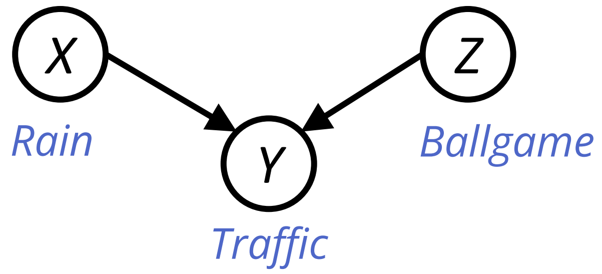


- Is X independent of Z?
  - ***No!*** Suffices to show one counterexample set of CPTs for which X is not independent of Z
  - E.g., homework due causes both forums and office hours to be busy
- Is X independent of Z given Y? ***Yes!***

$$P(z|x,y) = \frac{P(x,y,z)}{P(x,y)} = \frac{P(x|y)P(y)P(z|y)}{P(x|y)P(y)} = P(z|y)$$

Observing the cause blocks influence between effects

Common effect

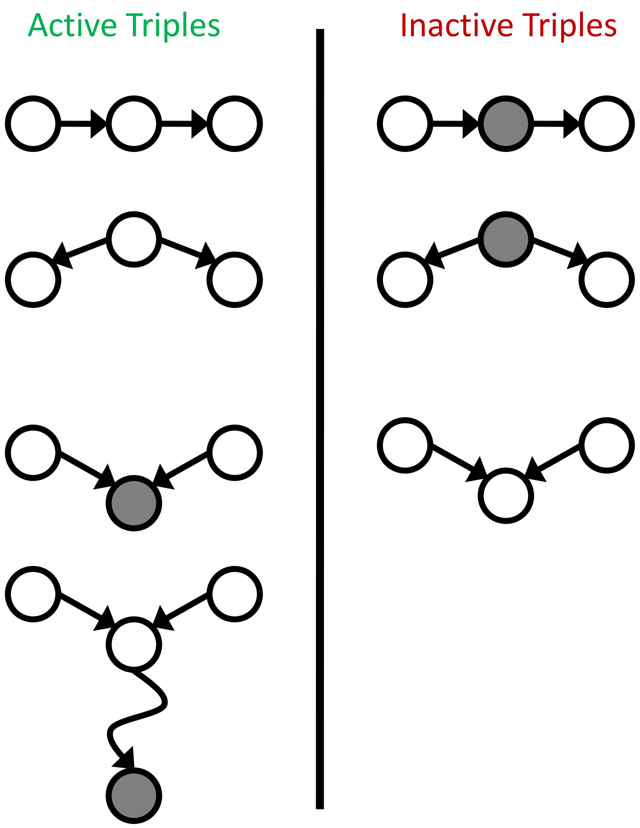


- Is X independent of Z?
  - ***Yes!*** The ballgame and the rain causes traffic, but they are not correlated
  - *Exercise: prove that they must be independent*
- Is X independent of Z given Y?
  - ***No!*** Seeing traffic puts the rain and the ballgame in competition as explanation

*Observing an effect **activates** influence between possible causes*
(this is backwards from the other two cases)

D-separation (textbook algorithm)

- In a given BN structure, are variables X and Y independent given evidence variables {Z}?
- Recipe: analyze the graph for *active paths*
  1. Shade evidence nodes
  2. Consider all undirected paths from X to Y
  3. Independence if no active paths (i.e., X and Y are “d-separated” by Z). A path is active if all triples in it are active.
- A path is active if each triple is active:
  - Causal chain (X→Y→Z) and common cause (X←Y→Z): if Y is unobserved
  - Common effect (X→Y←Z): if Y *or one of its descendants* is observed

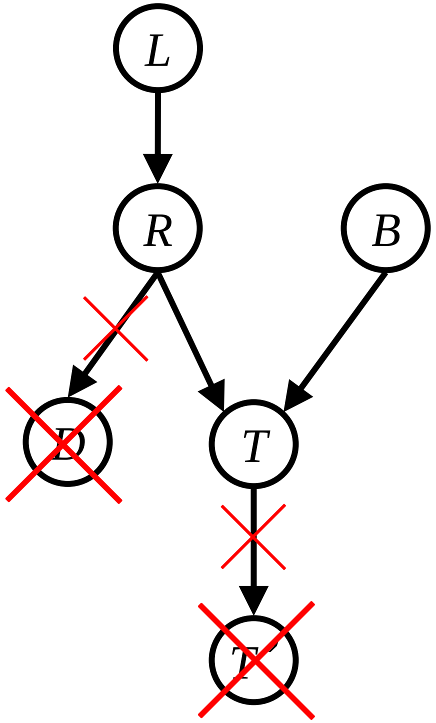


D-separation: graphical test

- In a given BN structure, are variables X and Y independent given evidence variables {Z}?
- i.e., Are X and Y d-separated by {Z}?
- Graphical test:
  1. Delete any leaf node W from the DAG as long as W is not in X, Y, or {Z}
  2. Repeat Step 1 until no more nodes can be deleted
  3. Delete all edges outgoing from nodes in {Z}
  4. X and Y are d-separated (independent) given Z iff they are ***disconnected*** in the resulting DAG

Example

1. Delete leaf nodes not in X, Y, or {Z}
2. Repeat Step 1 until no more can be deleted
3. Delete all outgoing edges from {Z}
4. $X \perp Y \mid Z$ if X and Y are ***disconnected*** in the resulting DAG

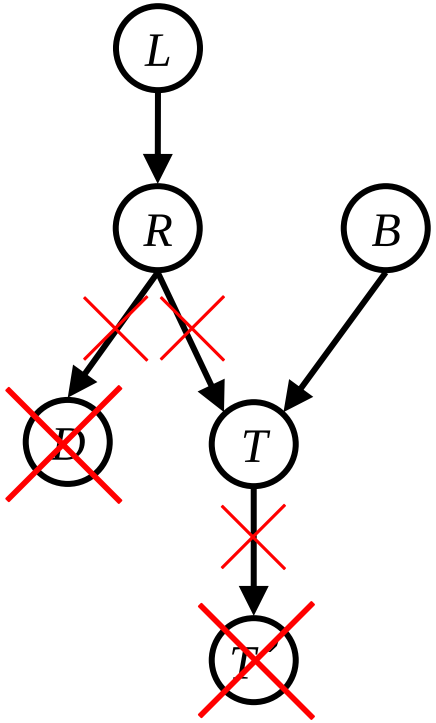


$L \perp B \mid T$?

Not d-separated

Example

1. Delete leaf nodes not in X, Y, or {Z}
2. Repeat Step 1 until no more can be deleted
3. Delete all outgoing edges from {Z}
4. $X \perp Y \mid Z$ if X and Y are ***disconnected*** in the resulting DAG

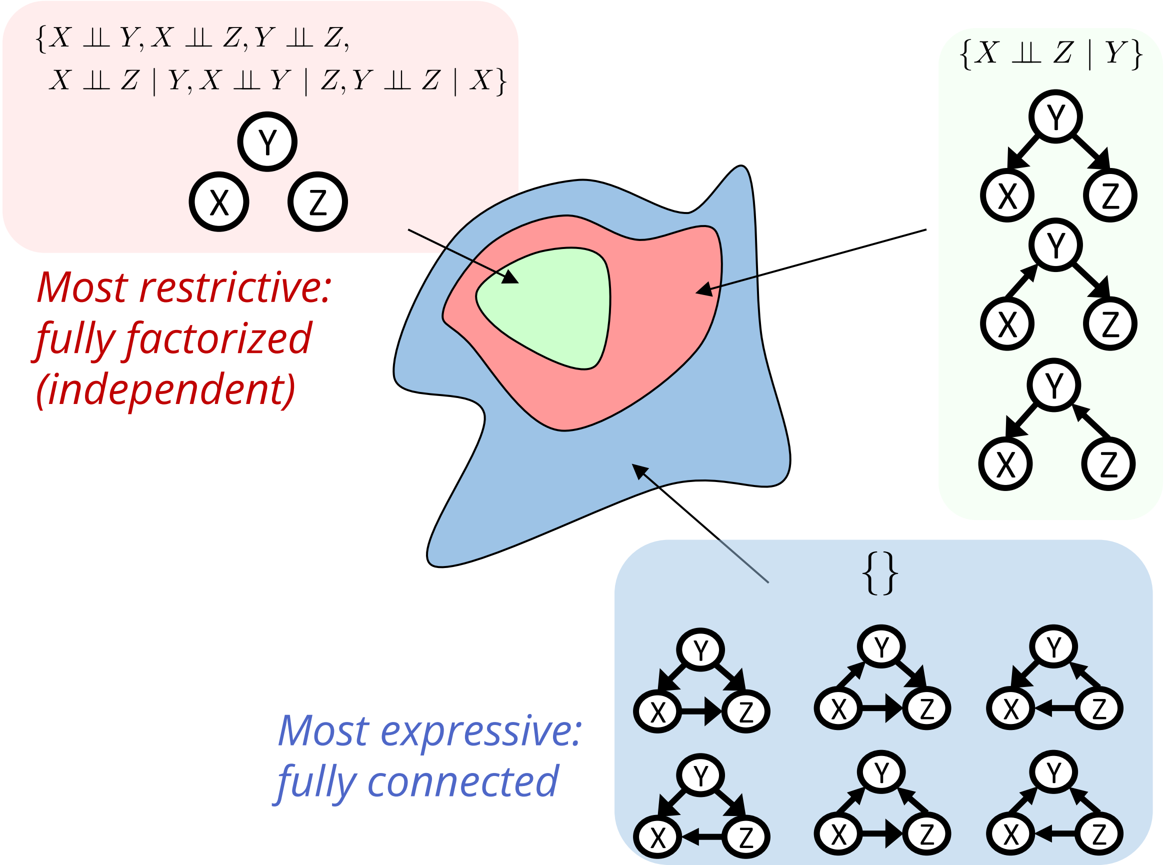


$L \perp B \mid T, R$?

L and B are d-separated given T, R

Topology limits distributions

- Given some graph topology G, only certain joint distributions can be encoded – *only those that satisfy (conditional) independences guaranteed by the graph.*
- Adding edges increases the set of possible distributions: *any independence not assumed by graph can be enforced by parameters.*
- However, adding edges increases the number of parameters and inference complexity



Summary

- Bayesian networks defined by structure (graph) and parameters (CPTs)
- Bayesian networks compactly represent joint distributions
- BN graph structure (topology) guarantees certain independencies of distributions
- D-separation gives precise conditional independence guarantees from graph alone
- BN joint distribution may still have further (conditional) independences that is not detectable until you inspect its specific distribution (including CPTs)
- Next: probabilistic inference algorithms on BNs
- Reading: R&N Ch.13